Table of Contents

..... 1

```
clear; clc; close all;

% Design Parameters
Fs = 10000;         % Sampling frequency in Hz
Ap = 0.5;           % Passband ripple in dB
As = 30;            % Stopband attenuation in dB

% 1. Low-Pass Filter Design
fp_lp = 1500;       % Passband frequency (Hz)
fs_lp = 2000;       % Stopband frequency (Hz)
wp_lp = fp_lp / (Fs/2);
ws_lp = fs_lp / (Fs/2);

[N_lp, Wn_lp] = cheb1ord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = cheby1(N_lp, Ap, Wn_lp, 'low');

% 2. High-Pass Filter Design
fp_hp = 2500;       % Passband frequency (Hz)
fs_hp = 1800;       % Stopband frequency (Hz)
wp_hp = fp_hp / (Fs/2);
ws_hp = fs_hp / (Fs/2);

[N_hp, Wn_hp] = cheb1ord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = cheby1(N_hp, Ap, Wn_hp, 'high');

% ========================================
% DISPLAY NUMERICAL OUTPUTS IN COMMAND WINDOW
% ========================================
fprintf('=== EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS ===\n\n');

fprintf('--- CHEBYSHEV LOW-PASS FILTER ---\n');
fprintf('Calculated Filter Order (N_lp): %d\n', N_lp);
fprintf('Passband Edge Cutoff Frequency (Wn_lp): %.4f Hz\n', Wn_lp * (Fs/2));
fprintf('Numerator Coefficients (b_lp):\n');
disp(b_lp);
fprintf('Denominator Coefficients (a_lp):\n');
disp(a_lp);

fprintf('--- CHEBYSHEV HIGH-PASS FILTER ---\n');
fprintf('Calculated Filter Order (N_hp): %d\n', N_hp);
fprintf('Passband Edge Cutoff Frequency (Wn_hp): %.4f Hz\n', Wn_hp * (Fs/2));
fprintf('Numerator Coefficients (b_hp):\n');
disp(b_hp);
fprintf('Denominator Coefficients (a_hp):\n');
disp(a_hp);

% Frequency Response Calculation
```

```
[H_lp, f] = freqz(b_lp, a_lp, 1024, Fs);
[H_hp, ~] = freqz(b_hp, a_hp, 1024, Fs);

% Plotting
figure('Name', 'Chebyshev Type-I LPF and HPF Responses');
subplot(2,2,1);
plot(f, 20*log10(abs(H_lp)), 'b', 'LineWidth', 1.5); grid on;
title(['Chebyshev LPF Magnitude (Order N = ', num2str(N_lp), ')']);
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)'); ylim([-60 5]);

subplot(2,2,2);
plot(f, angle(H_lp)*180/pi, 'r', 'LineWidth', 1.5); grid on;
title('Chebyshev LPF Phase Response');
xlabel('Frequency (Hz)'); ylabel('Phase (degrees)');

subplot(2,2,3);
plot(f, 20*log10(abs(H_hp)), 'b', 'LineWidth', 1.5); grid on;
title(['Chebyshev HPF Magnitude (Order N = ', num2str(N_hp), ')']);
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)'); ylim([-60 5]);

subplot(2,2,4);
plot(f, angle(H_hp)*180/pi, 'r', 'LineWidth', 1.5); grid on;
title('Chebyshev HPF Phase Response');
xlabel('Frequency (Hz)'); ylabel('Phase (degrees)');
```

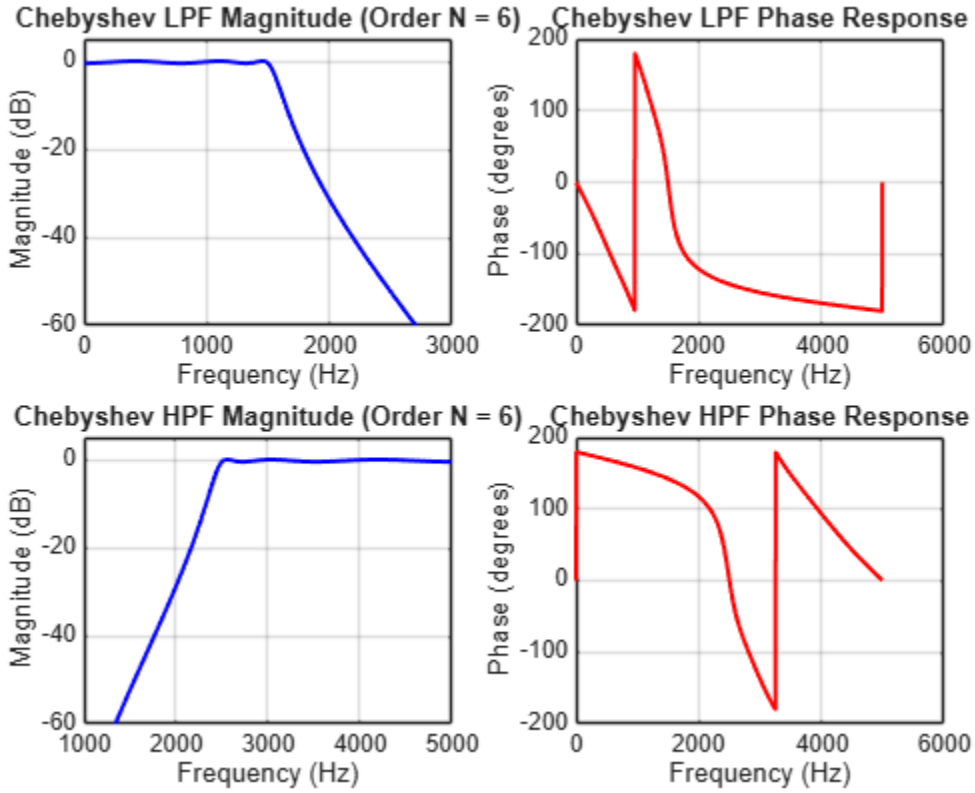
```
=== EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS ===

--- CHEBYSHEV LOW-PASS FILTER ---
Calculated Filter Order (N_lp): 6
Passband Edge Cutoff Frequency (Wn_lp): 1500.0000 Hz
Numerator Coefficients (b_lp):
    0.0006    0.0038    0.0095    0.0127    0.0095    0.0038    0.0006

Denominator Coefficients (a_lp):
    1.0000   -3.7650    6.8198   -7.3274    4.8717   -1.8927    0.3368

--- CHEBYSHEV HIGH-PASS FILTER ---
Calculated Filter Order (N_hp): 6
Passband Edge Cutoff Frequency (Wn_hp): 2500.0000 Hz
Numerator Coefficients (b_hp):
    0.0117   -0.0704    0.1761   -0.2348    0.1761   -0.0704    0.0117

Denominator Coefficients (a_hp):
    1.0000    1.3568    2.1347    1.8511    1.2978    0.5937    0.1650
```



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